



RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF APPLIED SCIENCES

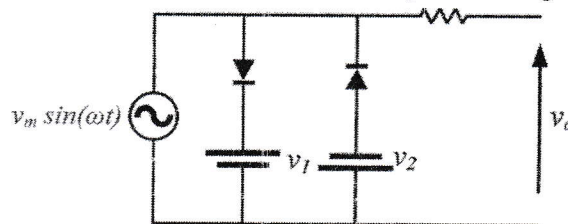
Bachelor of Science in Information Technology
First Year - Semester I Examination – February 2024

ICT 1303 – BASIC ELECTRONICS AND DIGITAL LOGIC DESIGN

Time: Three (03) hours

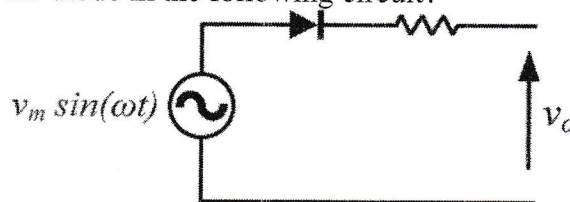
- There are **SIX (06)** questions in **THREE (03)** pages.
- All questions carry equal marks.
- Answer **ANY FIVE (05)** questions.

1. a) Explain what semiconductors are and why they cannot be used in constructing electronic devices in their pure form. (05 marks)
- b) Why do diodes conduct electricity only in one direction? (05 marks)
- c) Explain the operation of the following circuit using sketches of input and output waveforms. Assume that $V_1, V_2 < V_m$, where V_m is the peak voltage of AC source.



(05 marks)

- d) What is the function of the diode in the following circuit?



(05 marks)

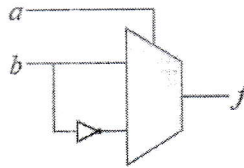
2. a) Discuss why the operation of bipolar junction transistors are sensitive to temperature. (05 marks)
- b) Why is it important to have a low output impedance for an amplifier? (05 marks)
- c) Explain how a depletion MOSFET differs from an enhancement MOSFET. (05 marks)
- d) Explain how a two-input NAND gate can be constructed using three MOSFETs. (05 marks)

3. a) Minimize the following Boolean expression and draw the minimum cost circuit.

$$f(a, b, c, d) = a\bar{d} + \bar{a}\bar{c}\bar{d} + \bar{a}c\bar{d} + \bar{a}d$$

(10 marks)

- b) The following circuit diagram shows a logic circuit implemented using a 2-to-1 multiplexer. Show that this circuit can be used to complement bit "b" using bit "a".

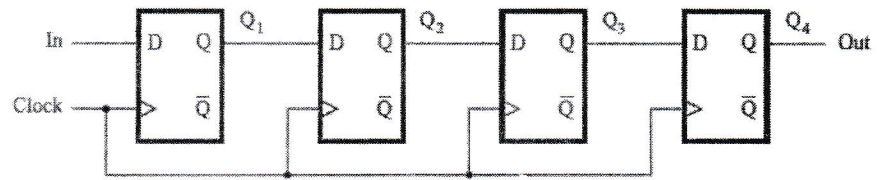


(10 marks)

4. A production system has four boilers to heat up paint of four colours namely, red, yellow, blue, and white. During the production any two colours from red, yellow, and blue are mixed. White is not mixed with any color. The process starts when the appropriate boilers are hot. Four sensors indicate the temperature of the four boilers. Design the electronic circuit that signals the start of the process. You are required to write down the truth table, minterms and the Boolean expression. Then minimize the expression and draw the minimal circuit. Clearly state all assumptions you made. (20 marks)

5. a) Distinguish between shift registers and parallel registers. (05 marks)

- b) Explain the operation of the following shift register using timing diagrams.



(10 marks)

- c) Why cannot level triggered flip-flops be used in designing of sequential circuits?

(05 marks)

6. Design a decimal (modulo-10) synchronous up/down counter using D flip-flops. (20 marks)

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